

```
In [1]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.linear_model import LinearRegression
        import matplotlib.pyplot as plt
```

```
In [7]: data = pd.read_csv('Body_weight.csv')
```

```
In [9]: data.head()
```

```
Out[9]:
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

```
In [11]: data.shape
```

```
Out[11]: (10, 4)
```

```
In [13]: data.isnull().sum()
```

```
Out[13]: Weight    0
         Height    0
         Age       0
         Gender    0
         dtype: int64
```

```
In [15]: data['Gender'] = data['Gender'].map({'Male':1, 'Female':0})
```

```
In [17]: x = data.iloc[:,1:4].values
         y = data.iloc[:,0:1].values
```

```
In [19]: x_train, x_test, y_train, y_test = train_test_split(x,y,
         test_size=0.2, random_state = 42)
```

```
In [21]: model = LinearRegression()
         model.fit(x_train, y_train)
         y_pred = model.predict(x_test)
```

```
In [23]: print(y_pred)
         print(y_test)
```

```
[[56.85651025]
 [79.10576115]]
[[64]
 [69]]
```

```
In [29]: print(model.intercept_)
```

```
[-3.05521724]
```

```
In [31]: print(model.coef_)
```

```
[[32.8353363  0.31729474 14.62311848]]
```

```
In [35]: from sklearn.metrics import r2_score, mean_squared_error
```

```
In [37]: print("R2 Score :", r2_score(y_test, y_pred))
```

```
R2 Score : -11.252468343750815
```

```
In [39]: print("Mean Squared Error :", mean_squared_error(y_test, y_pred))
```

```
Mean Squared Error : 76.5779271484426
```

```
In [41]: new_person = [[1.75,30,1]]
prediction = model.predict(new_person)
print("Predicted Weight =", prediction)
```

```
Predicted Weight = [[78.54858202]]
```

```
In [43]: import numpy as np
```

```
idx = np.argsort(x_train[:,0])
```

```
plt.scatter(x_train[:,0], y_train, color='blue')
```

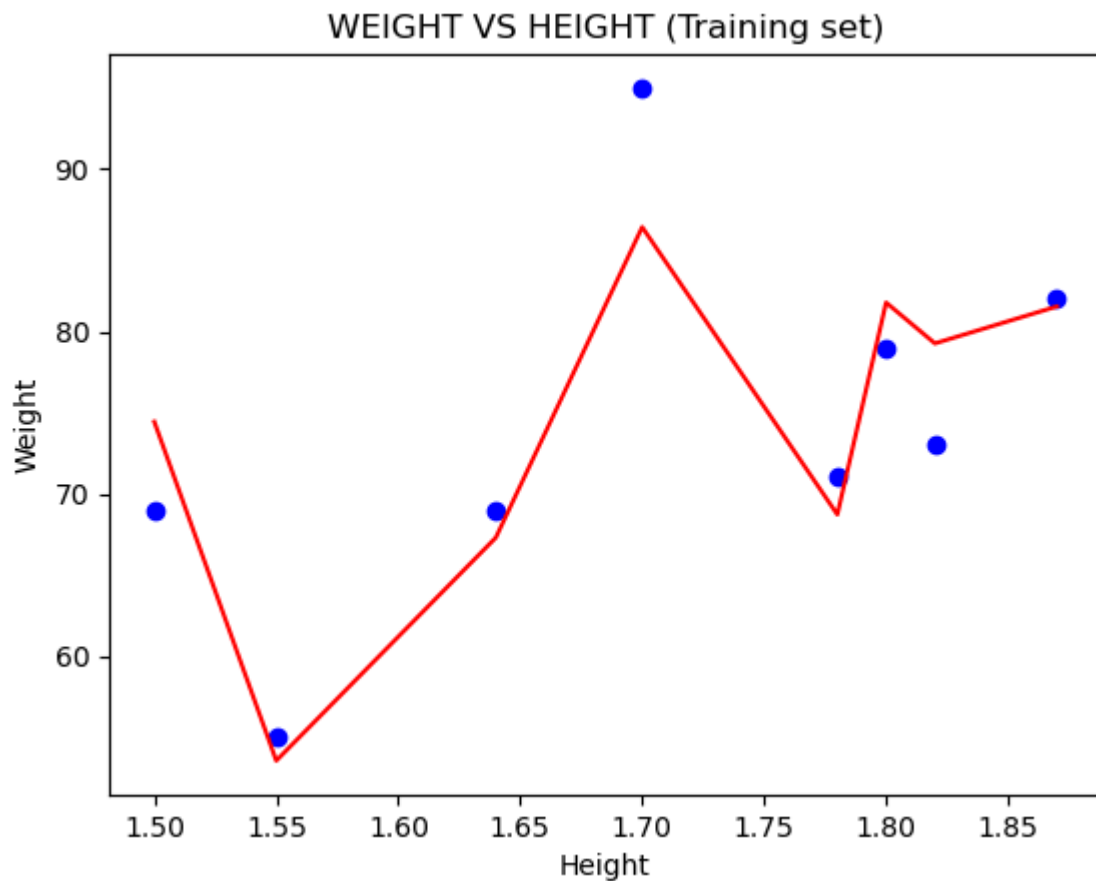
```
plt.plot(x_train[idx,0], model.predict(x_train)[idx], color='red')
```

```
plt.title('WEIGHT VS HEIGHT (Training set)')
```

```
plt.xlabel('Height')
```

```
plt.ylabel('Weight')
```

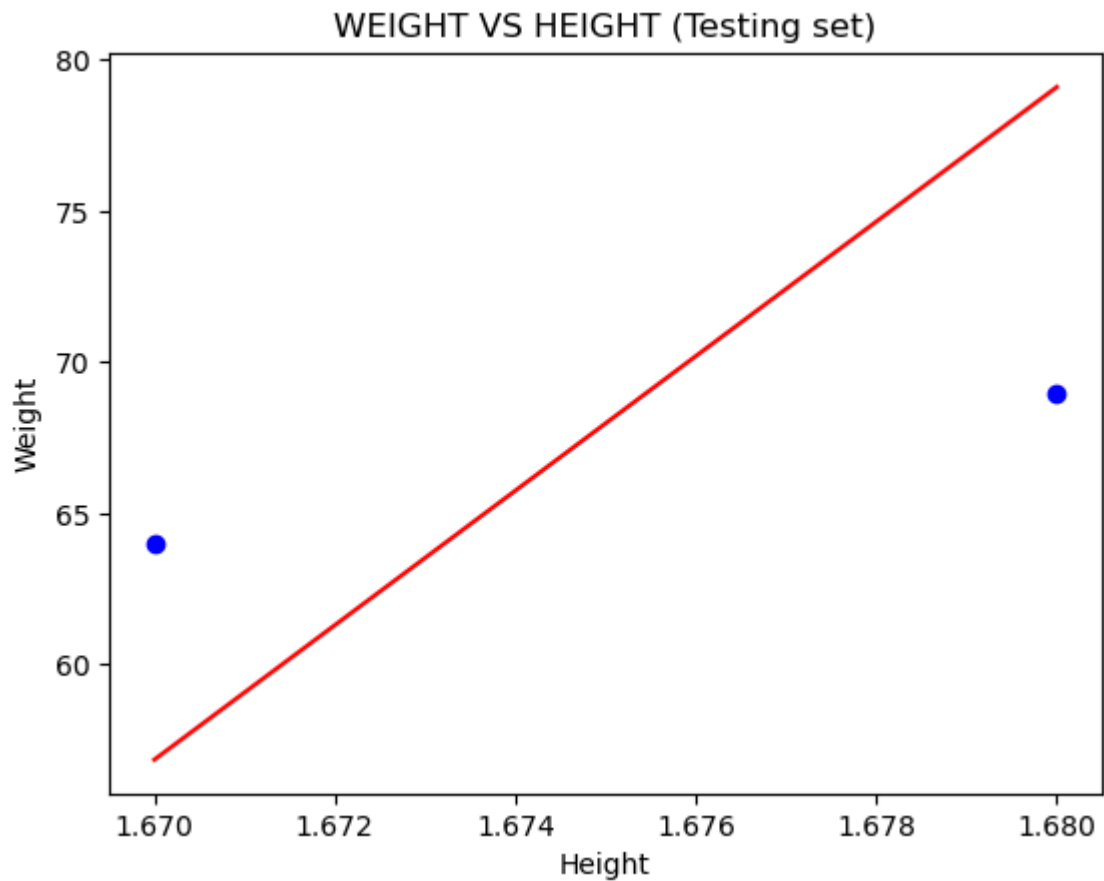
```
plt.show()
```



```
In [45]: idx = np.argsort(x_test[:,0])

plt.scatter(x_test[:,0], y_test, color='blue')
plt.plot(x_test[idx,0], model.predict(x_test)[idx], color='red')

plt.title('WEIGHT VS HEIGHT (Testing set)')
plt.xlabel('Height')
plt.ylabel('Weight')
plt.show()
```



In []: